

Lab 2 – Wireshark Network Traffic Analysis

Cybersecurity Homelab Portfolio – Network Traffic Investigation

Objective

To capture and analyse real network traffic using Wireshark, identify DNS resolution, observe a TCP three-way handshake, and examine TLS 1.3 connection establishment from a SOC analyst perspective.

1. Environment

Tool	Wireshark
Capture Interface	Ethernet
Local IPv4 observed	192.168.1.6
Local IPv6 observed	2402:d000:810c:2b25:4c48:cb2:eeb7:a329
Protocols	DNS, UDP, TCP, TLS 1.3, IPv4 and IPv6

2. Investigation Procedure

Wireshark was opened on the Ethernet interface and network traffic was captured. Display filters were used to isolate DNS, TCP and TLS traffic, and individual packets were examined to understand the communication sequence.

3. DNS Investigation

A DNS query for **chatgpt.com** was identified. The query originated from **192.168.1.6** and was sent to **192.168.1.1** using UDP destination port 53. The query used DNS record type **HTTPS** (type 65).

The corresponding response contained two **AAAA** records:

```
2a06:98c1:3101::6812:202f
2606:4700:4408::ac40:9bd1
```

This demonstrated that a domain can resolve to multiple IPv6 addresses and that DNS can return different record types.

4. TCP Three-Way Handshake

A complete TCP three-way handshake was identified for an IPv6 connection between the local host **2402:d000:810c:2b25:4c48:cb2:eeb7:a329** and remote host **2603:1063:2001:1d3a::365:ff1** using port 443.

Step	Direction	Packet	Purpose
1	57462 → 443	SYN	Client requests a TCP connection
2	443 → 57462	SYN, ACK	Server acknowledges and responds
3	57462 → 443	ACK	Client acknowledges; connection established

The handshake confirmed successful TCP connection establishment before application-layer secure communication proceeded.

5. TLS 1.3 Investigation

Following the TCP handshake, a TLS 1.3 **Client Hello** was observed from the local IPv6 host to the remote server. The packet contained the Server Name Indication (SNI) **editor.svc.cloud.microsoft**.

A subsequent server response contained **Server Hello, Application Data**. This indicates that TLS negotiation progressed and that application traffic was protected by TLS encryption.

Because TLS 1.3 encrypts application-layer content, the protected communication was not directly readable in the packet capture. The SNI provided the hostname requested by the client during TLS establishment.

6. Filters Used

DNS: dns

TCP: tcp

TLS: tls

Connection filter: tcp.port == 57462

7. Findings

Finding	Observation
DNS Resolution	A DNS query for chatgpt.com and a response containing two AAAA IPv6 records were observed.
TCP Connectivity	A complete SYN → SYN-ACK → ACK sequence was observed for an IPv6 connection to port 443.
TLS Security	TLS 1.3 negotiation was observed after TCP establishment.
SNI	The TLS Client Hello identified editor.svc.cloud.microsoft as the requested hostname.
Encryption	Application Data was observed as encrypted TLS traffic rather than readable application content.
IPv4 / IPv6	Both IPv4 and IPv6 traffic were present in the capture.

8. SOC Analyst Interpretation

The traffic observed represents a normal sequence of network communication: DNS resolution, TCP connection establishment, TLS negotiation, and encrypted application data. No malicious activity was concluded from this capture. The exercise demonstrates the ability to identify protocol stages and distinguish connection establishment from encrypted application traffic.

9. Conclusion

The Wireshark investigation successfully demonstrated practical network traffic analysis. DNS packets were analysed to understand hostname resolution, a complete TCP three-way handshake was identified, and TLS 1.3 negotiation was examined through the Client Hello and Server Hello. The lab provided practical experience with packet filtering, IPv4/IPv6 traffic, TCP connection states, DNS records, TLS and encrypted application communication.

10. Evidence Collected

Recommended evidence to retain with this report:

- DNS query screenshot for chatgpt.com
- DNS response screenshot showing the AAAA records
- TCP SYN, SYN-ACK and ACK screenshots
- TLS 1.3 Client Hello screenshot showing the SNI
- TLS Server Hello / Application Data screenshot

- Original Wireshark .pcapng capture file